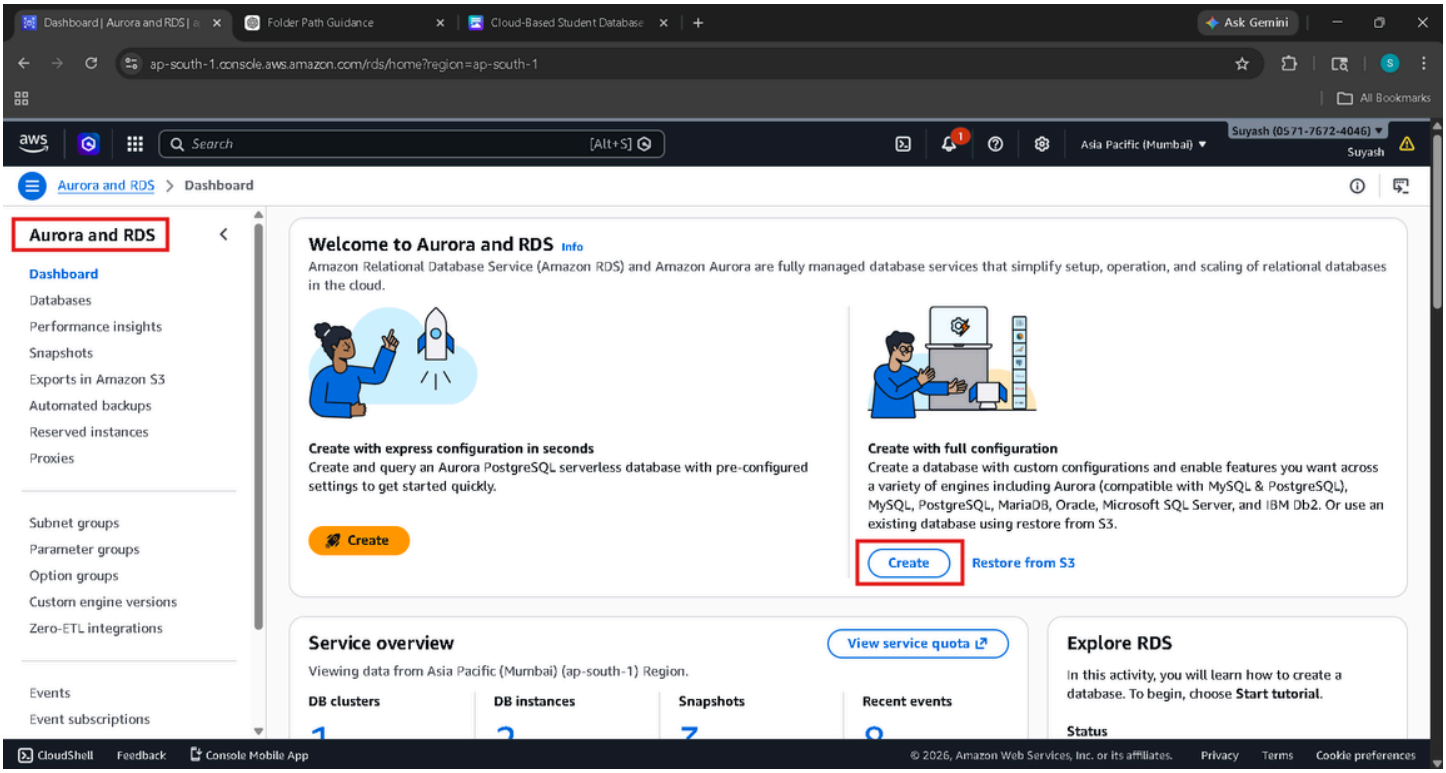
AWS RDS Student Management System

sign in to aws page

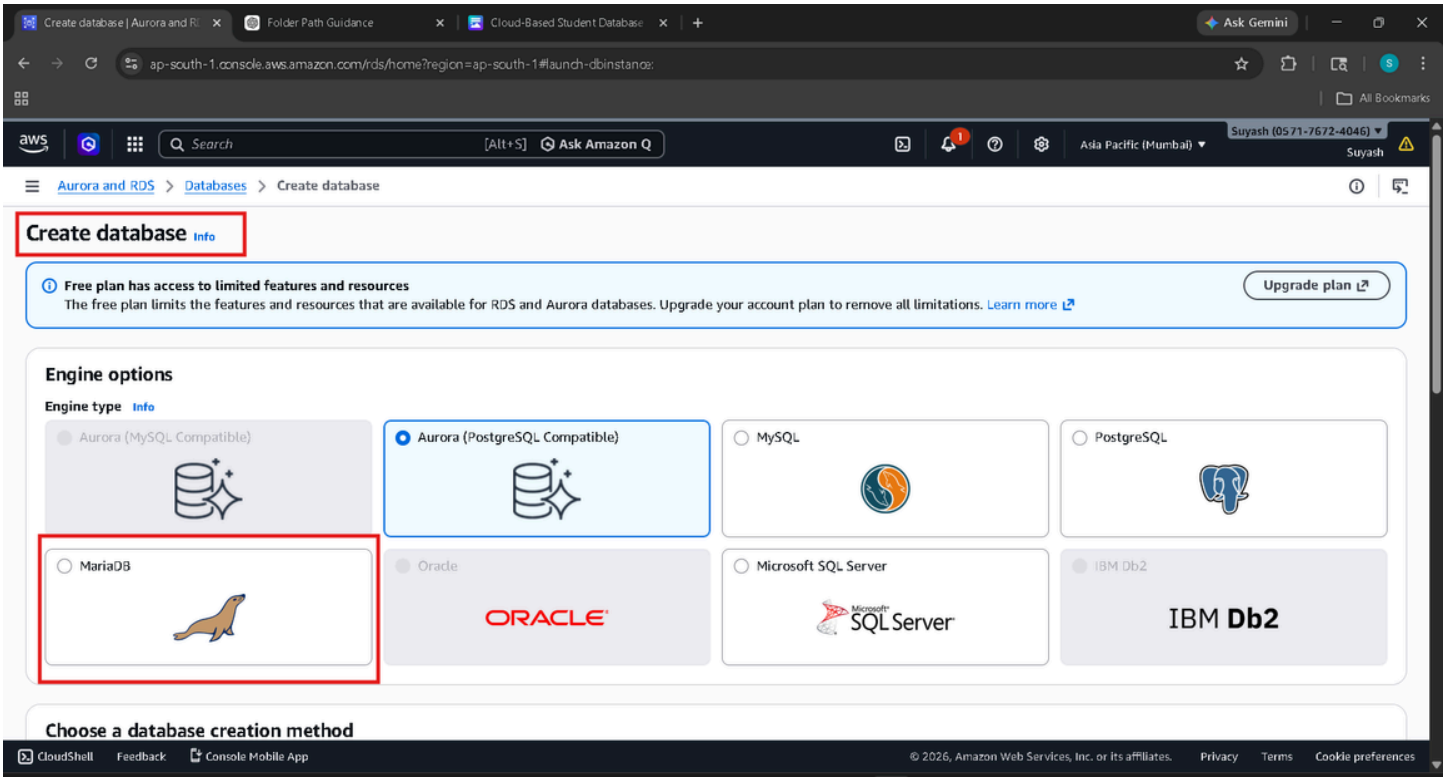
Go in RDS service

Click on Create RDS

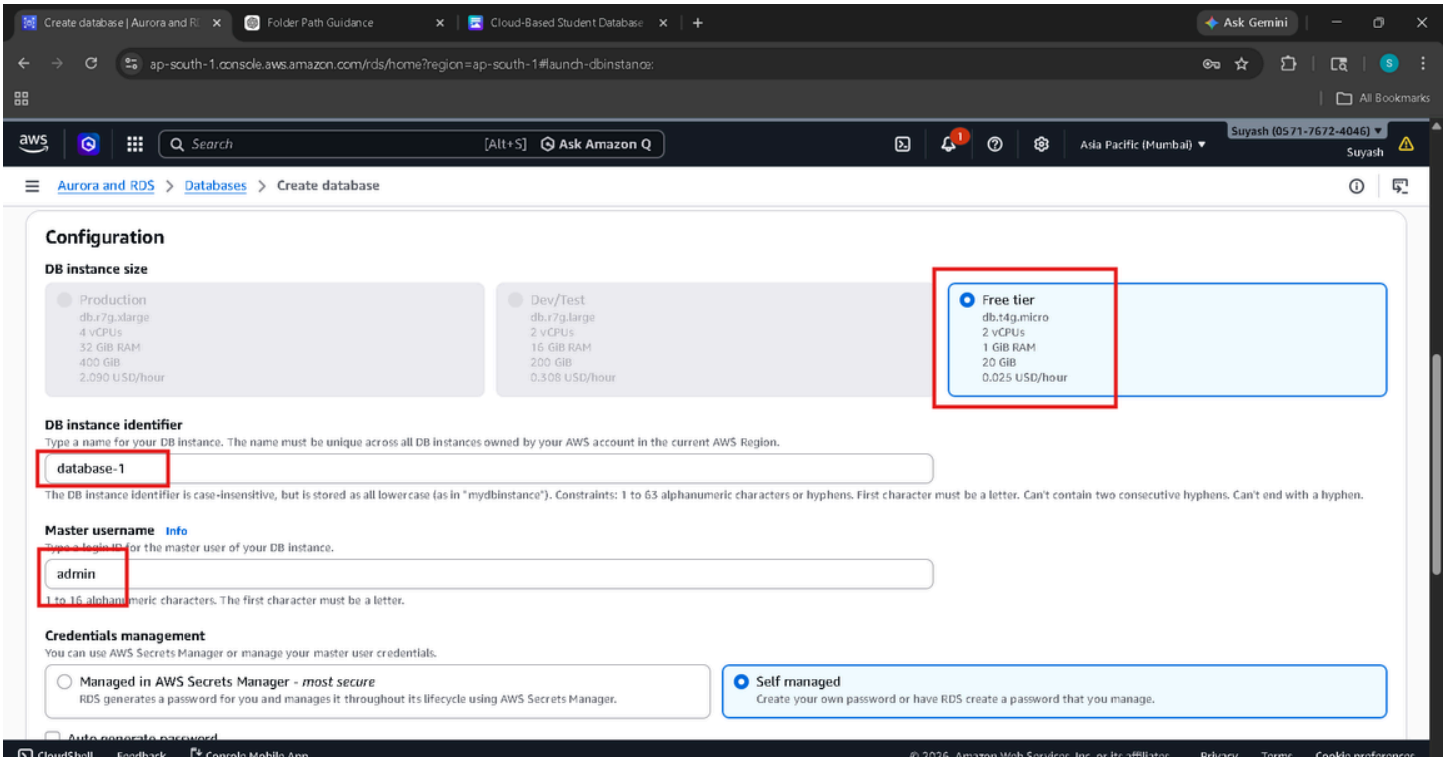


You enter in Create Database page.

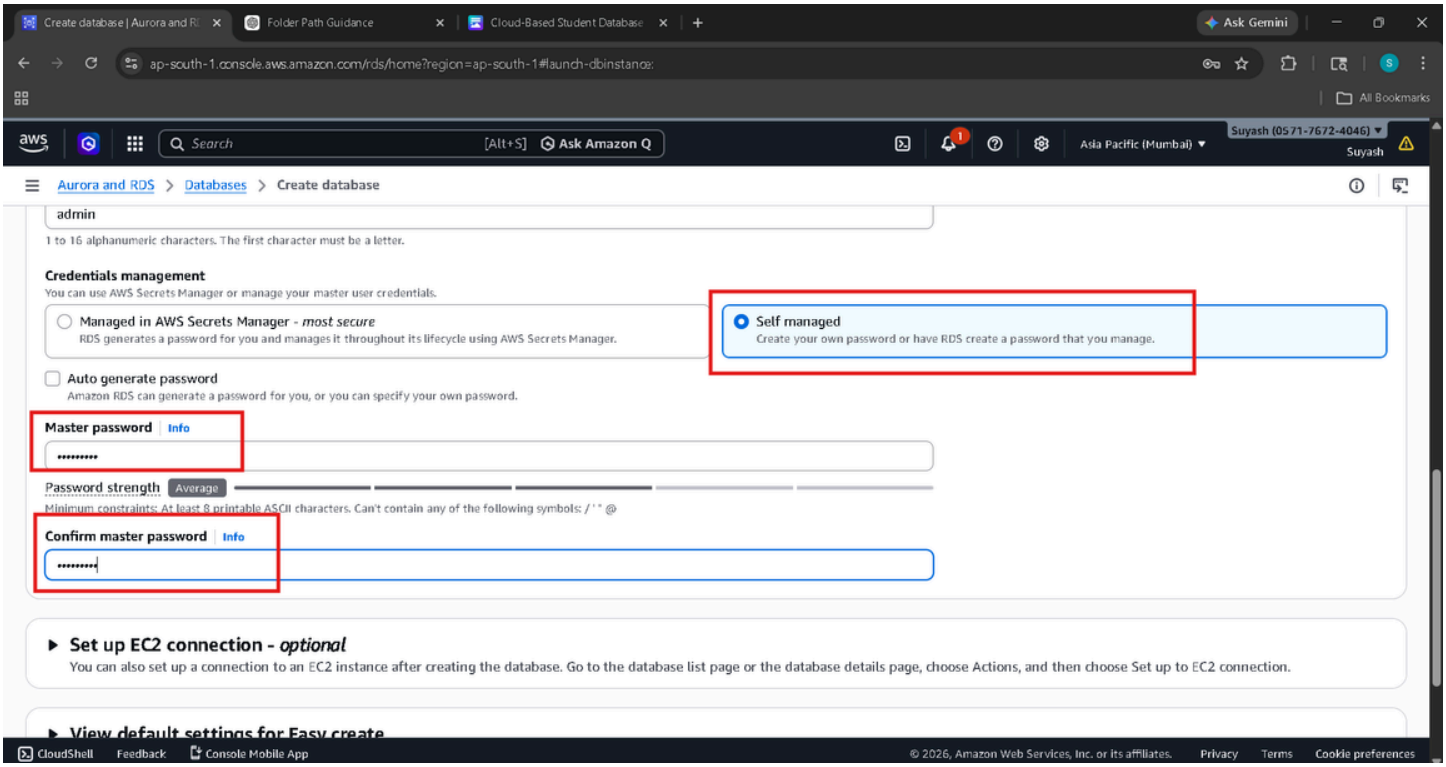
Select MariaDB For database



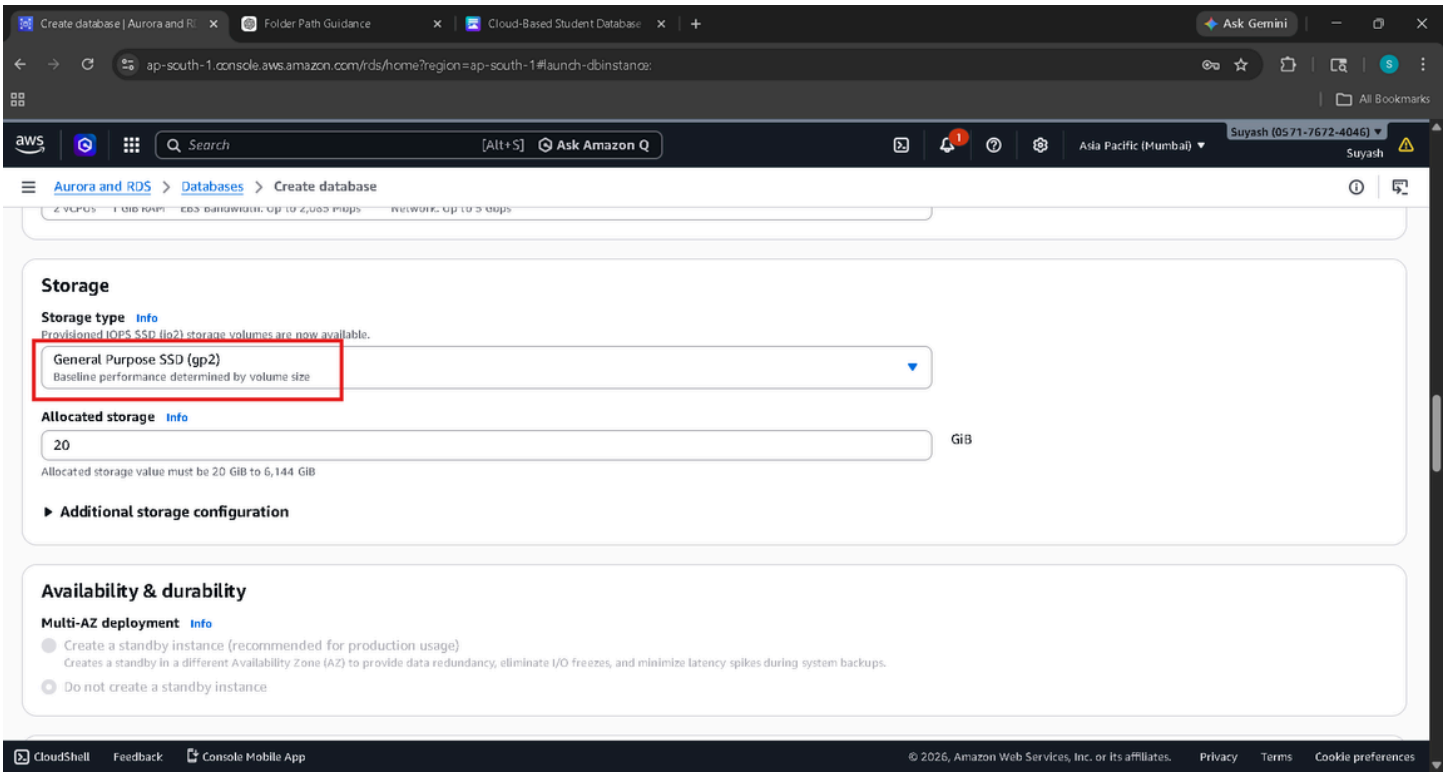
Put all as it is



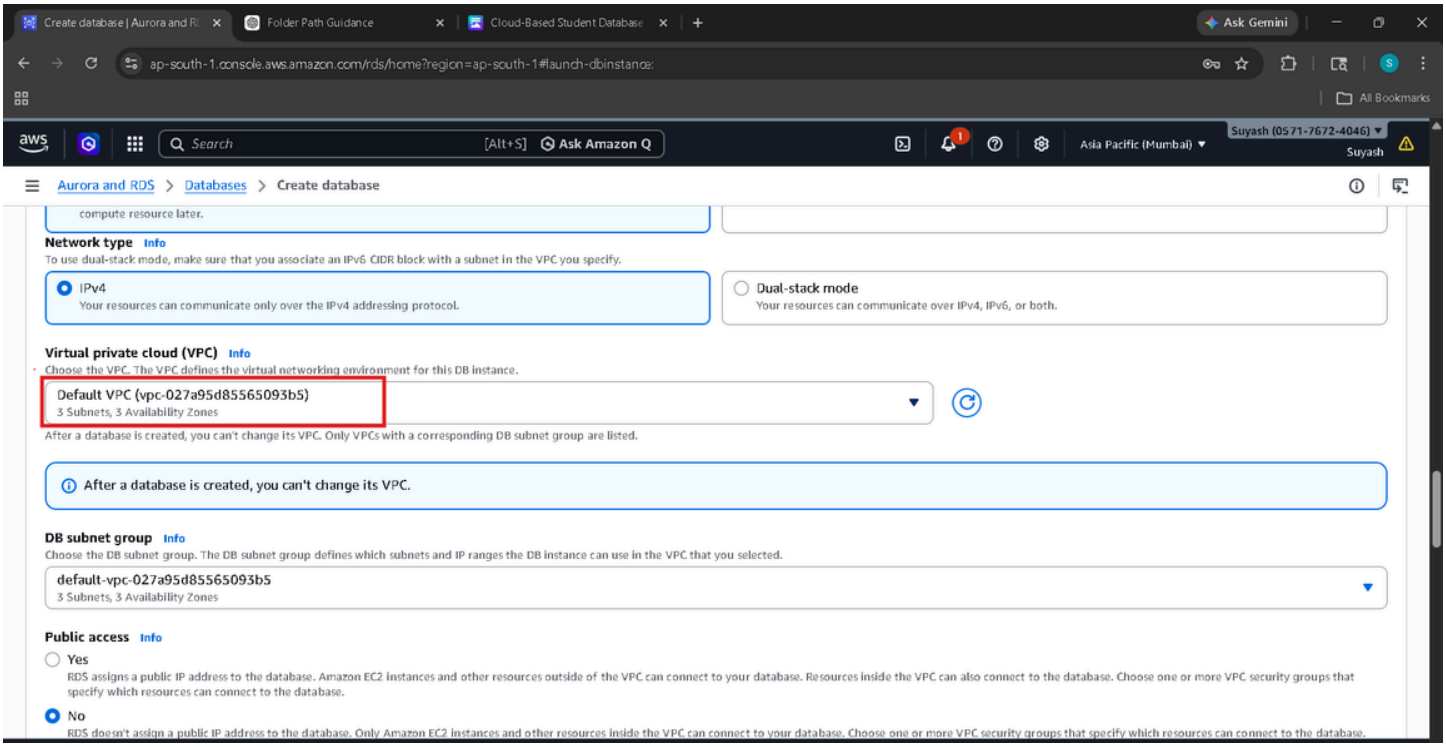
Select user and password for Self managed



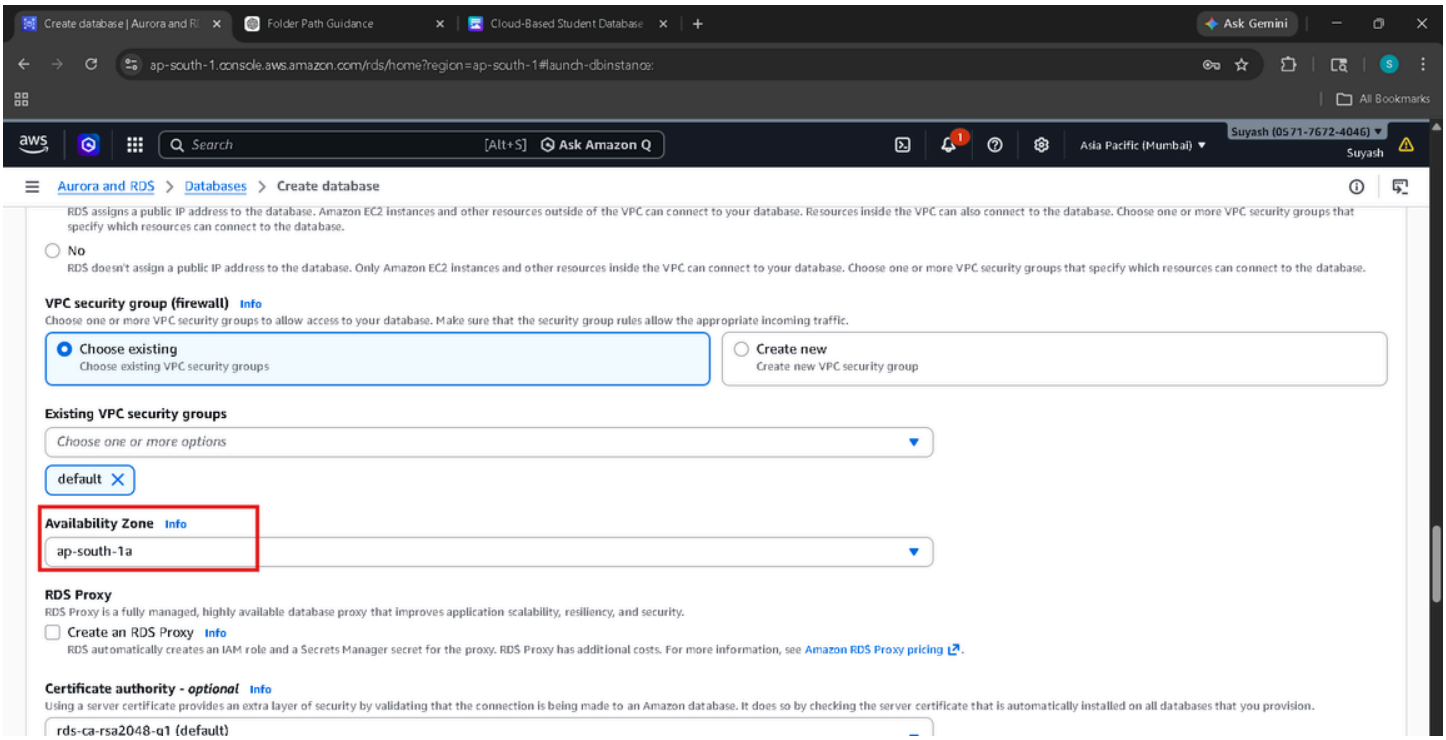
Storage



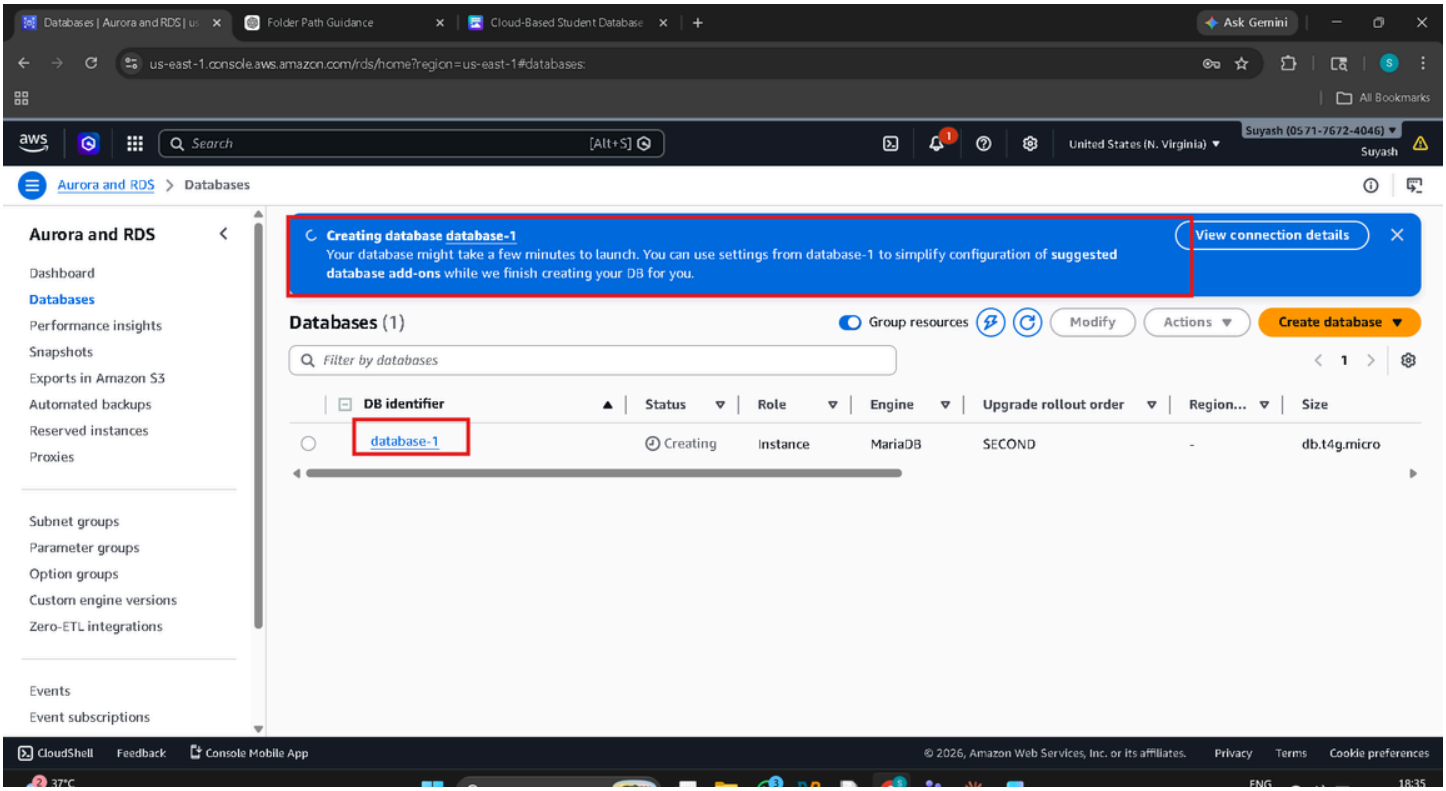
Put VPC as default



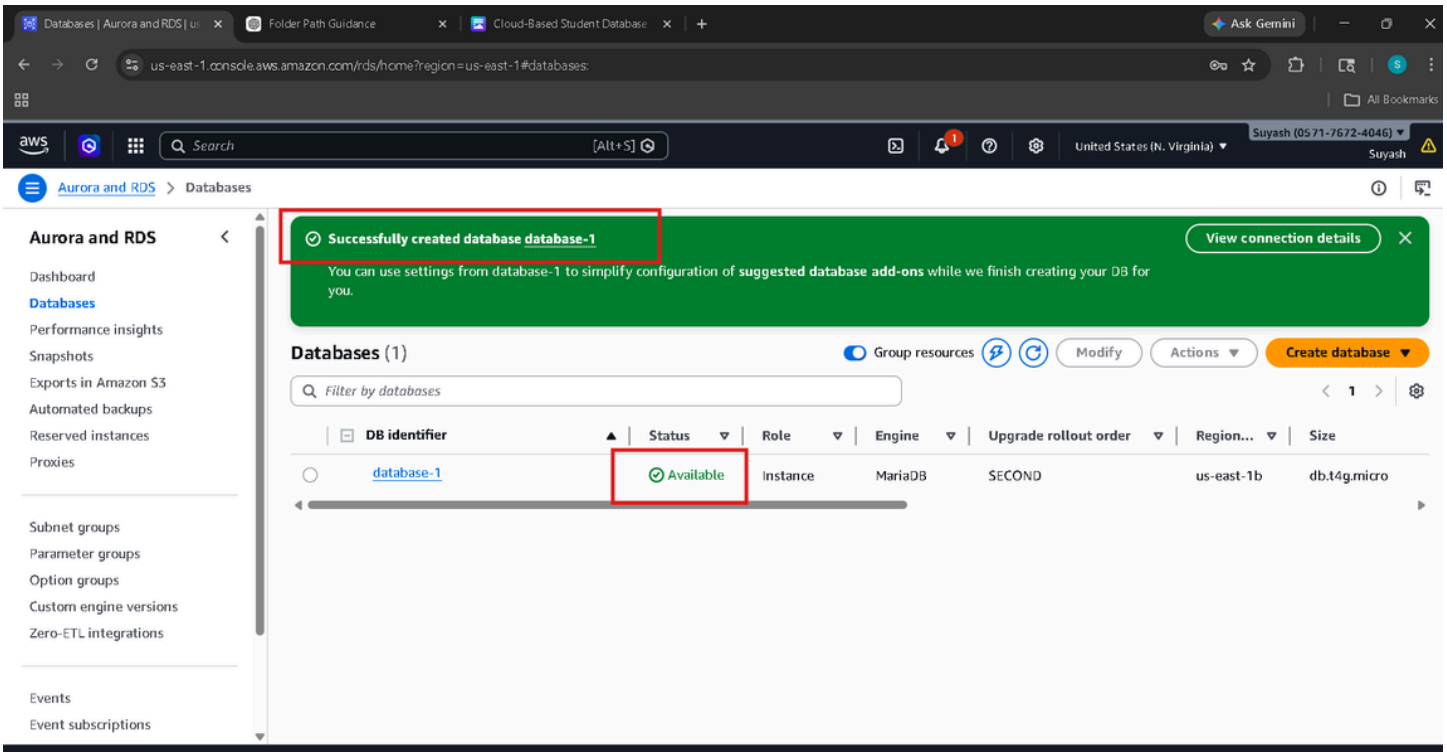
Chose A.Z. as ap-south-1a



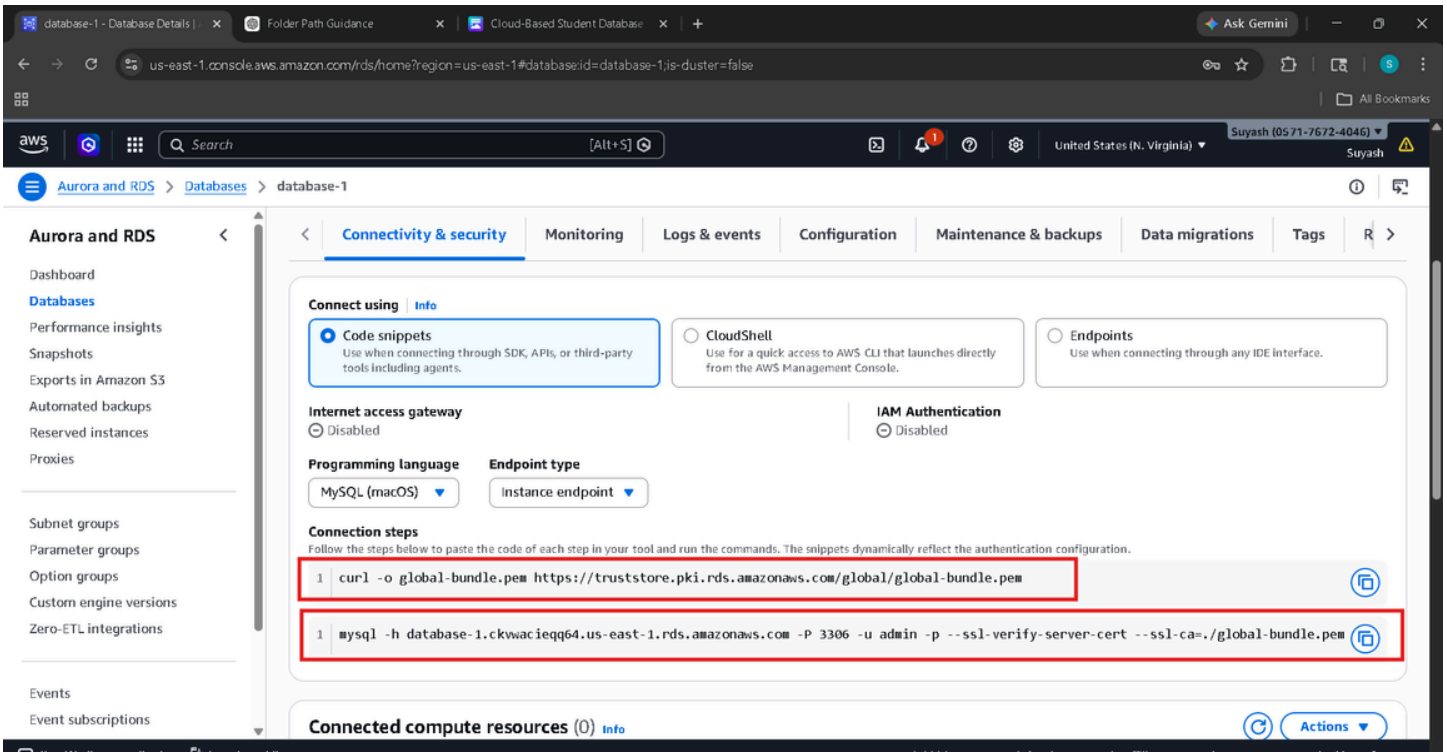
It will take some time to launch



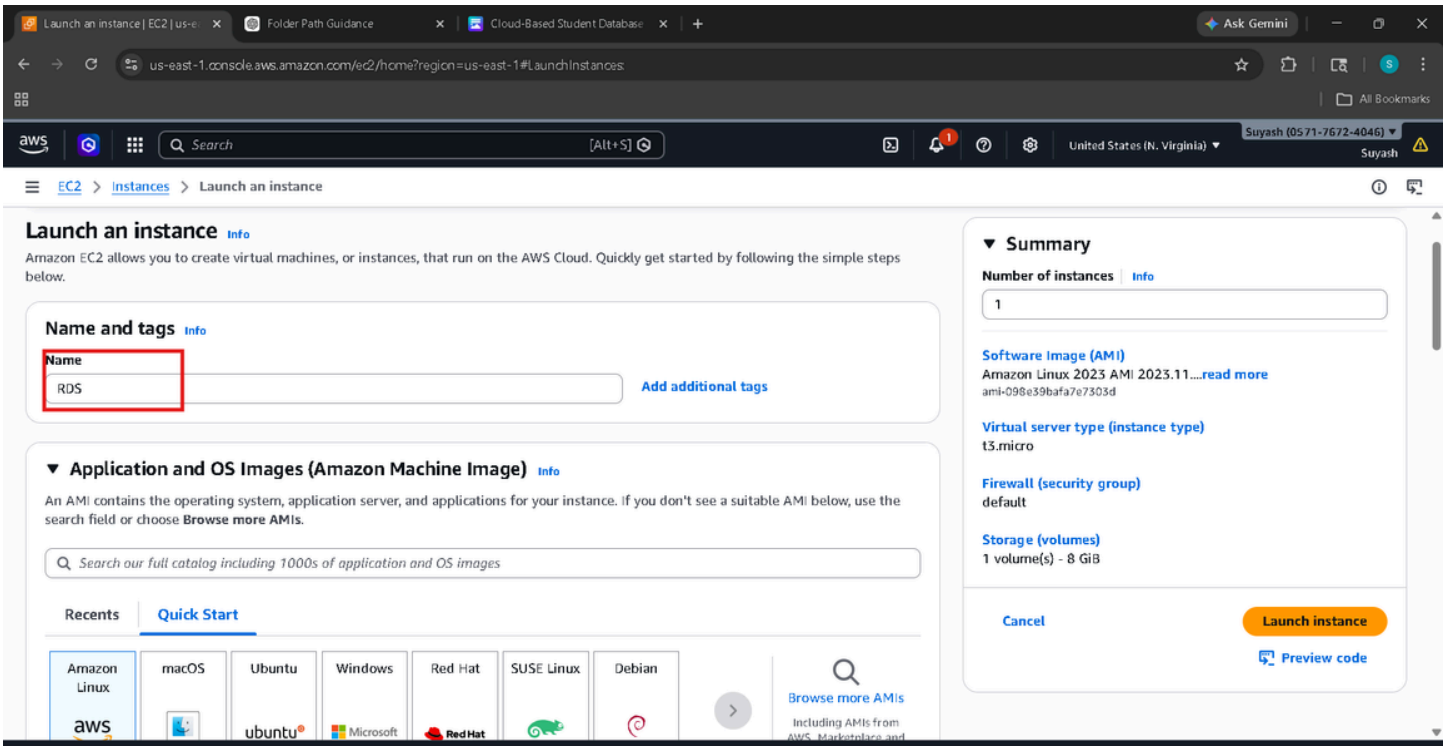
Now it's Available to use



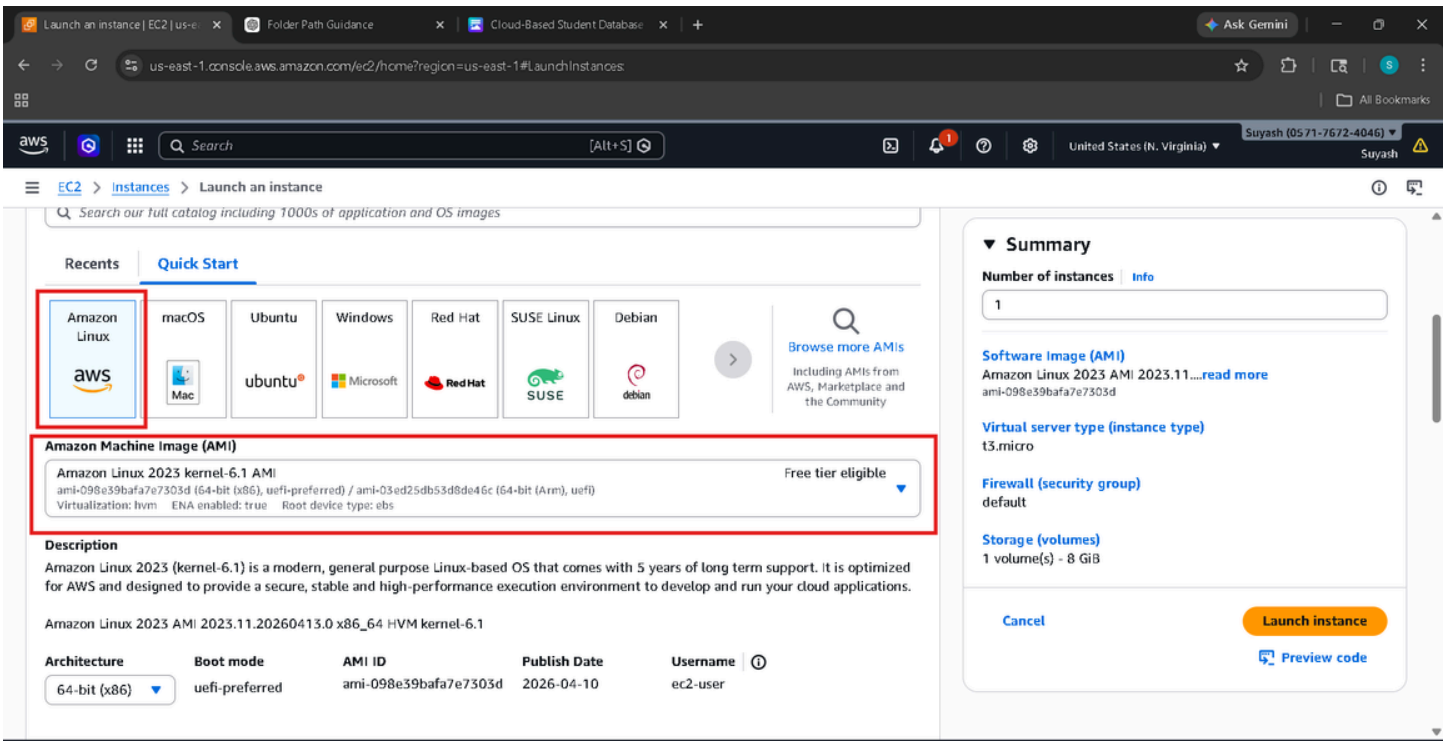
After created you will see the connection steps links



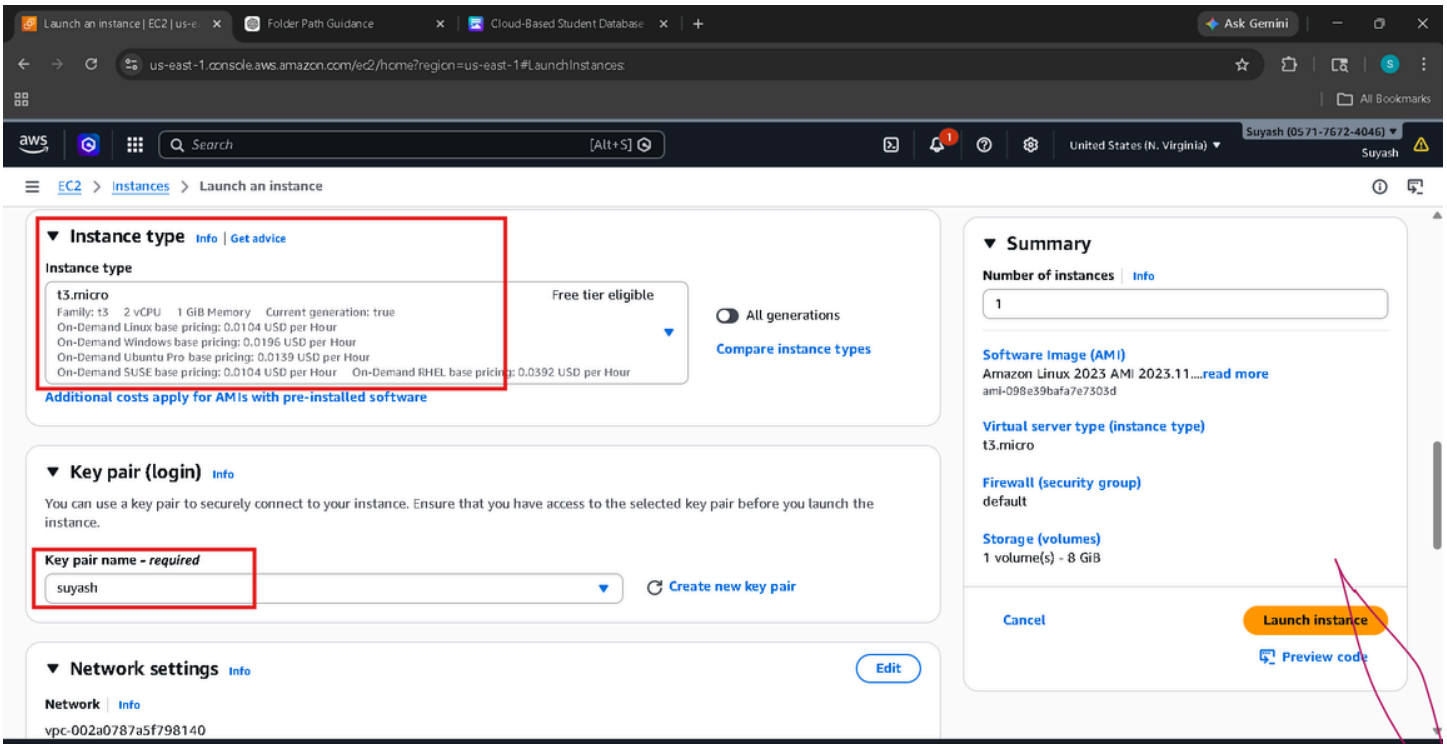
Now create an ec2 instance for database



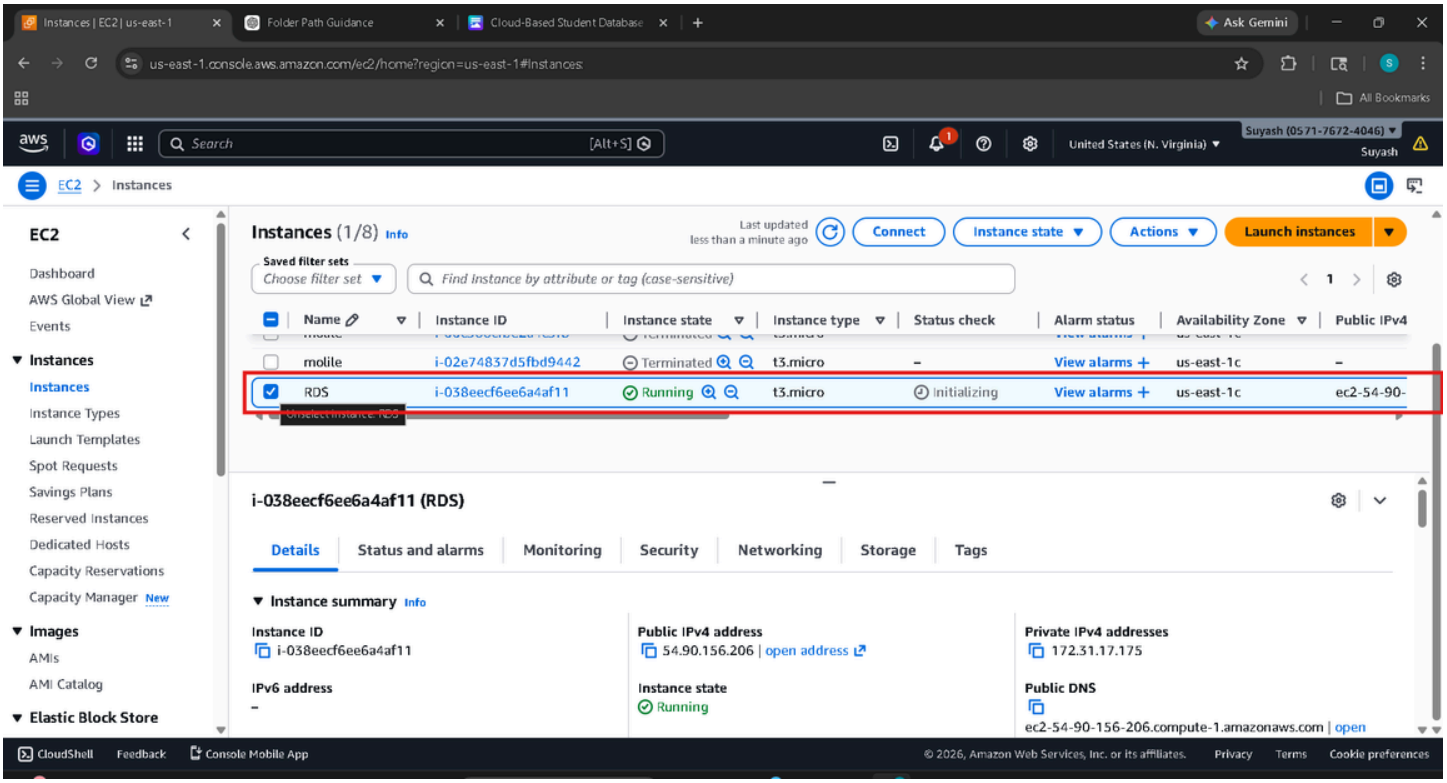
Select OS



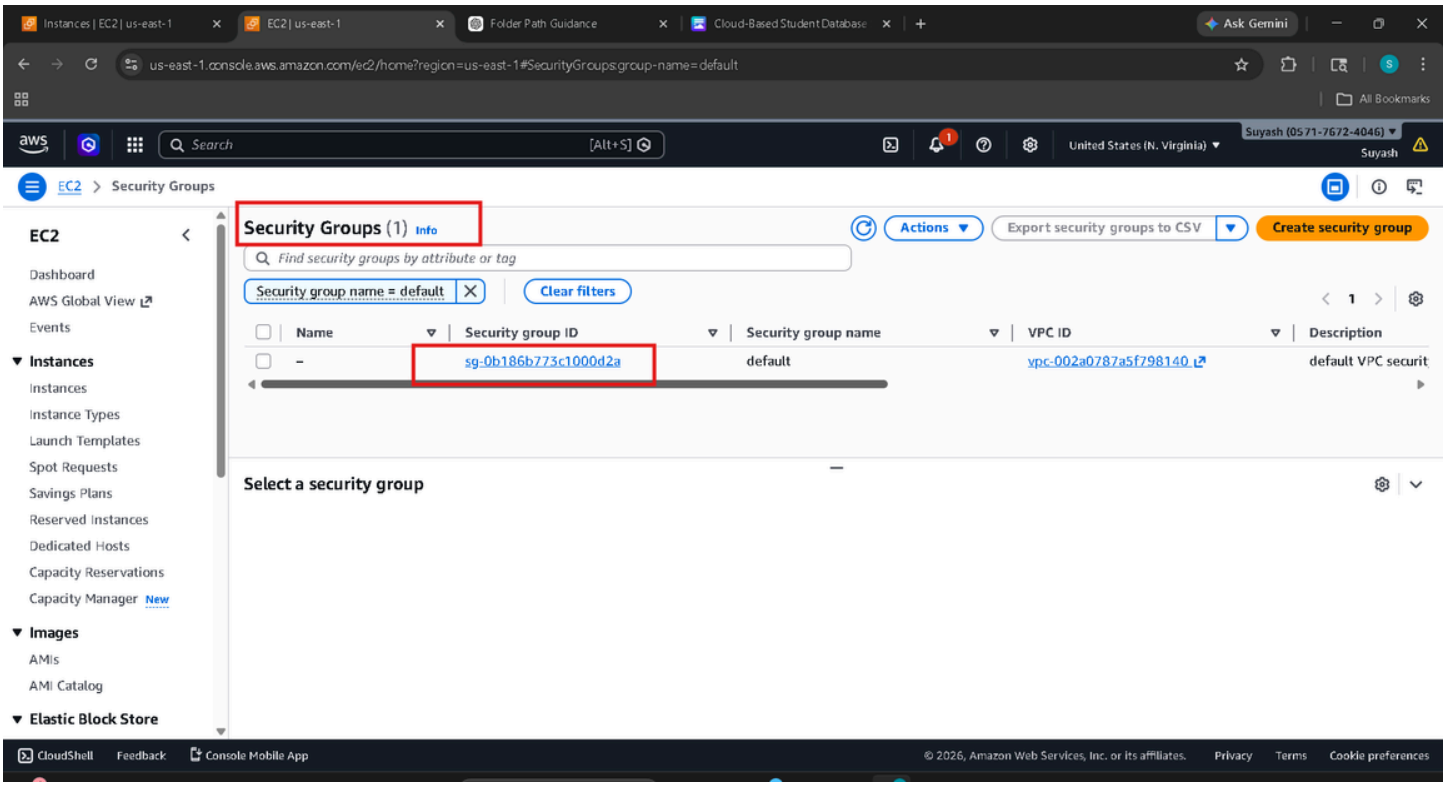
Instance type and key-pair



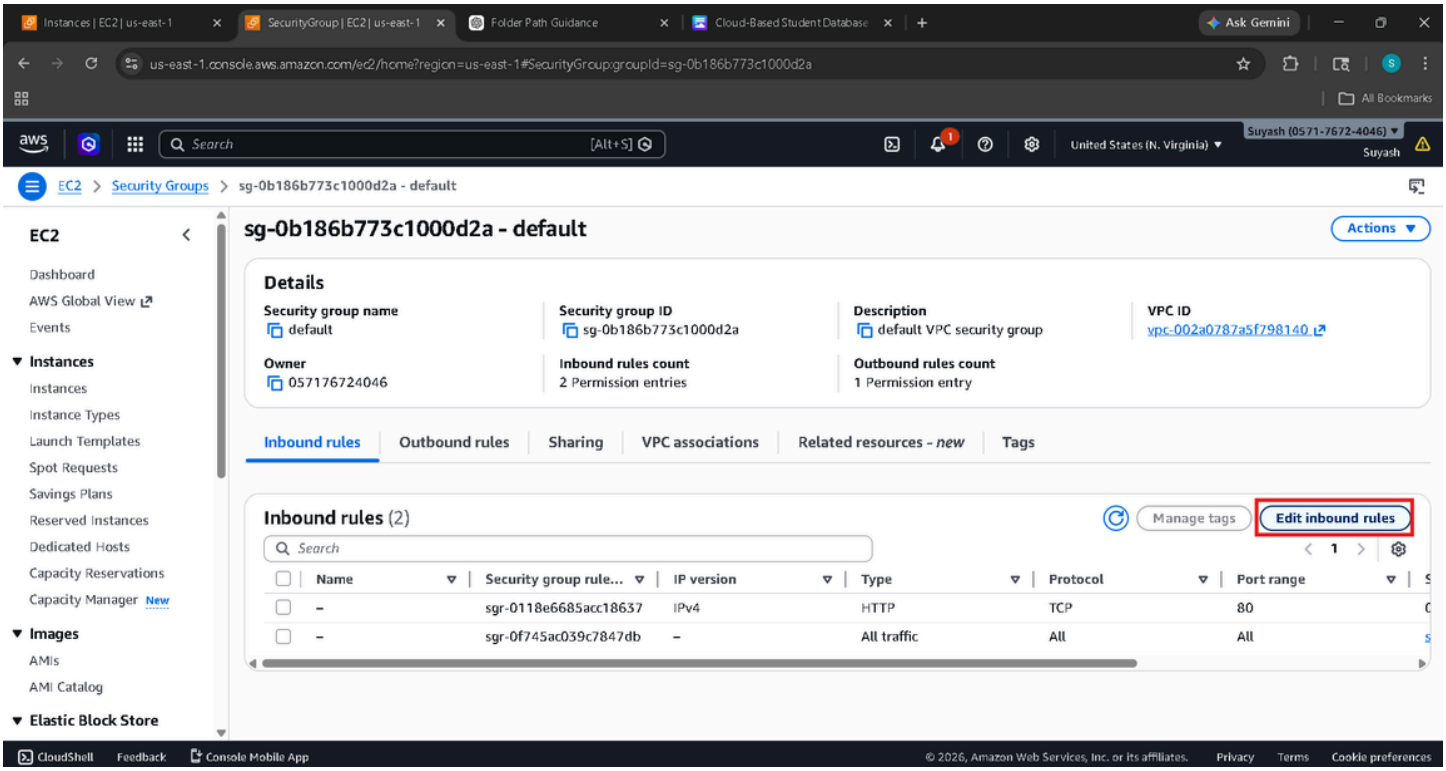
Now launch instance



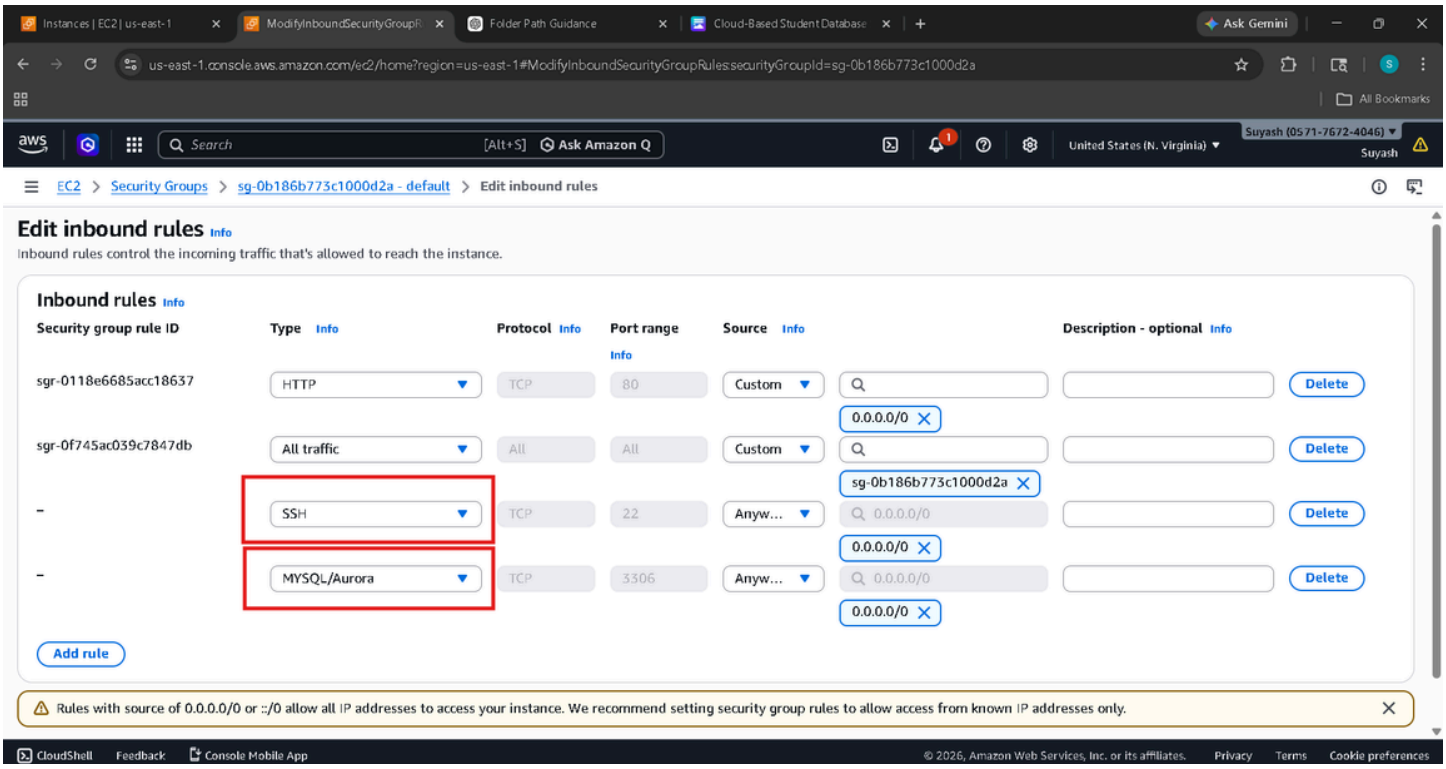
Go in security group



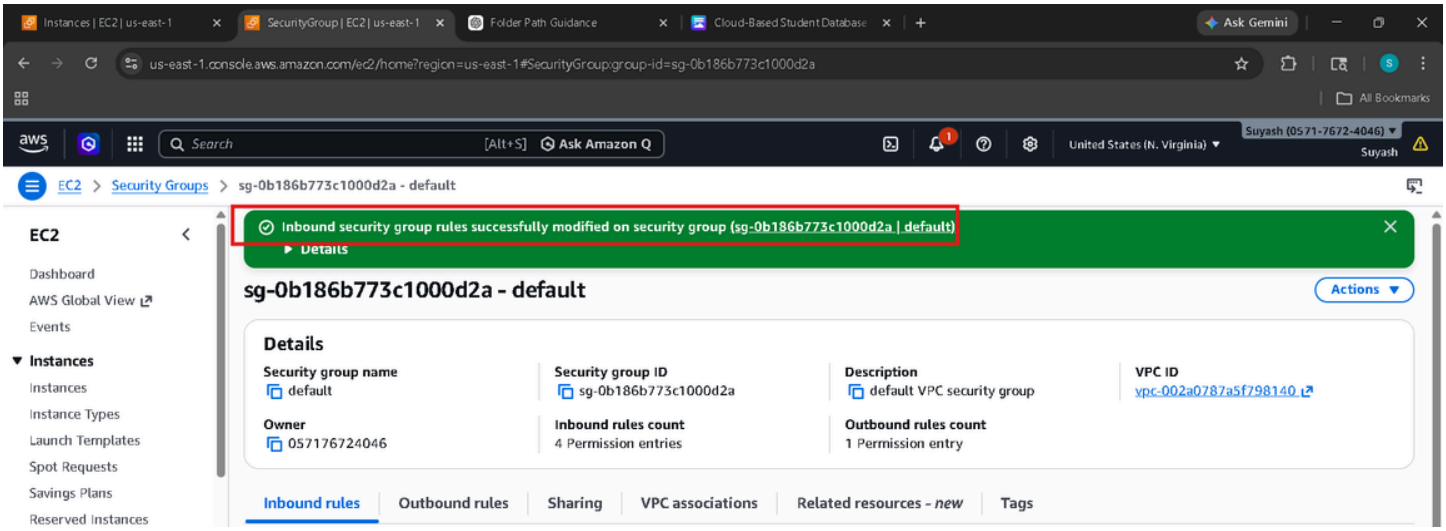
Click on Edit inbound rule



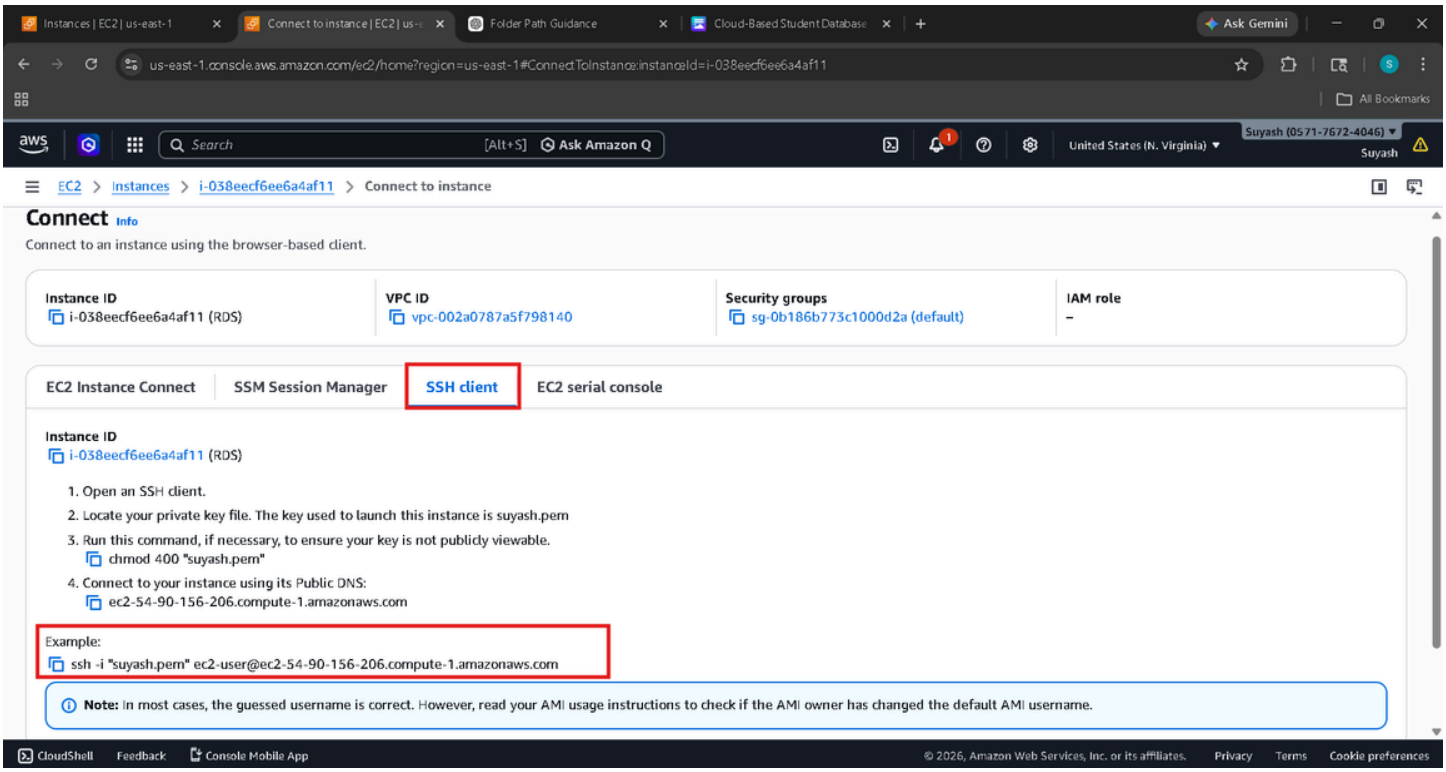
Add MYSQL and SSH in inbound rules.



Add Rule



go to connect instance and copy ssh form ssh client



GO on git bash and give path for your key pair

Command -”cd /d/crome\ downloads”

Now do to git bash paste your SSH link on it

ssh -i "suyash.pem" ec2-user@ec2-54-90-156-206.compute-1.amazonaws.com

you will get axis of instance.

```
suyas@Suyash MINGW64 /d/crome downloads
$ ssh -i "suyash.pem" ec2-user@ec2-54-90-156-206.compute-1.amazonaws.com
The authenticity of host 'ec2-54-90-156-206.compute-1.amazonaws.com (54.90.156.206)' can't be established.
ED25519 key fingerprint is SHA256:2Gt3OFVPK8nnLVgOJoovUvEg1zg62pJZW8TqSH+G5oA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-54-90-156-206.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
       #_
       ~\_  ####_        Amazon Linux 2023
      ~~  \_#####\
      ~~     \###|
      ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
       ~~       V~' '->
        ~~~         /
          ~~._.   _/
             _/ _/
           _/m/'
[ec2-user@ip-172-31-17-175 ~]$
```

type - sudo -i for enter in root user

```
             _/ _/
           _/m/'
[ec2-user@ip-172-31-17-175 ~]$ sudo -i
[root@ip-172-31-17-175 ~]#
```

Go on RDS and copy the connection step and paste on git bash

curl -o global-bundle.pem https://truststore.pki.rds.amazonaws.com/global/global-bundle.pem

```
suyas@Suyash MINGW64 /d/crome downloads
$ ssh -i "suyash.pem" ec2-user@ec2-54-90-156-206.compute-1.amazonaws.com
The authenticity of host 'ec2-54-90-156-206.compute-1.amazonaws.com (54.90.156.206)' can't be established.
ED25519 key fingerprint is SHA256:2Gt3OFVPK8nnLVgOJoovUvEg1zg62pJZW8TqSH+G5oA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-54-90-156-206.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
       #_
       ~\_  ####_        Amazon Linux 2023
      ~~  \_#####\
      ~~     \###|
      ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
       ~~       V~' '->
        ~~~         /
          ~~._.   _/
             _/ _/
           _/m/'
[ec2-user@ip-172-31-17-175 ~]$ sudo -i
[root@ip-172-31-17-175 ~]# curl -o global-bundle.pem https://truststore.pki.rds.amazonaws.com/global/global-bundle.pem
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 165408 100 165408    0     0  6411k      0  --:--:-- --:--:-- --:--:--  6730k
[root@ip-172-31-17-175 ~]#
[root@ip-172-31-17-175 ~]#
```

Install MariaDB in instance

commands-

yum install mariadb105-server -y

```
root@ip-172-31-17-175:~
================================================================================================================================================================================================================
 Package                                                    Architecture                    Version                                                    Repository                                                    Size
================================================================================================================================================================================================================
Installing:
 mariadb105-server                                          x86_64                          3:10.5.29-1.amzn2023.0.1                                   amazonlinux                                                   10 M
Installing dependencies:
 mariadb-connector-c                                        x86_64                          3.3.10-1.amzn2023.0.1                                      amazonlinux                                                  211 k
 mariadb-connector-c-config                                 noarch                          3.3.10-1.amzn2023.0.1                                      amazonlinux                                                  9.9 k
 mariadb105                                                 x86_64                          3:10.5.29-1.amzn2023.0.1                                   amazonlinux                                                  1.5 M
 mariadb105-common                                          x86_64                          3:10.5.29-1.amzn2023.0.1                                   amazonlinux                                                   28 k
 mariadb105-errmsg                                          x86_64                          3:10.5.29-1.amzn2023.0.1                                   amazonlinux                                                  212 k
 mysql-selinux                                              noarch                          1.0.4-2.amzn2023.0.3                                       amazonlinux                                                   36 k
 perl-DBD-MariaDB                                           x86_64                          1.22-1.amzn2023.0.4                                        amazonlinux                                                  153 k
 perl-DBI                                                   x86_64                          1.643-7.amzn2023.0.3                                       amazonlinux                                                  700 k
 perl-File-Copy                                             noarch                          2.34-477.amzn2023.0.7                                      amazonlinux                                                   20 k
 perl-Math-BigInt                                           noarch                          1:1.9998.39-2.amzn2023.0.2                                 amazonlinux                                                  202 k
 perl-Math-BigRat                                           noarch                          0.2624-500.amzn2023.0.2                                    amazonlinux                                                   42 k
 perl-Math-Complex                                          noarch                          1.59-477.amzn2023.0.7                                      amazonlinux                                                   46 k
 perl-Sys-Hostname                                          x86_64                          1.23-477.amzn2023.0.7                                      amazonlinux                                                   16 k
Installing weak dependencies:
 mariadb105-backup                                          x86_64                          3:10.5.29-1.amzn2023.0.1                                   amazonlinux                                                  6.0 M
 mariadb105-cracklib-password-check                         x86_64                          3:10.5.29-1.amzn2023.0.1                                   amazonlinux                                                   13 k
 mariadb105-gssapi-server                                   x86_64                          3:10.5.29-1.amzn2023.0.1                                   amazonlinux                                                   15 k
 mariadb105-server-utils                                    x86_64                          3:10.5.29-1.amzn2023.0.1                                   amazonlinux                                                  207 k

Transaction Summary
================================================================================================================================================================================================================
Install  18 Packages

Total download size: 19 M
Installed size: 116 M
Downloading Packages:
(1/18): mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64.rpm                                                                                                                   4.9 MB/s | 211 kB     00:00
(2/18): mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch.rpm                                                                                                            220 kB/s | 9.9 kB     00:00
(3/18): mariadb105-10.5.29-1.amzn2023.0.1.x86_64.rpm                                                                                                                            25 MB/s | 1.5 MB     00:00
(4/18): mariadb105-common-10.5.29-1.amzn2023.0.1.x86_64.rpm                                                                                                                     1.1 MB/s |  28 kB     00:00
(5/18): mariadb105-cracklib-password-check-10.5.29-1.amzn2023.0.1.x86_64.rpm                                                                                                   512 kB/s |  13 kB     00:00
(6/18): mariadb105-errmsg-10.5.29-1.amzn2023.0.1.x86_64.rpm                                                                                                                     7.3 MB/s | 212 kB     00:00
(7/18): mariadb105-backup-10.5.29-1.amzn2023.0.1.x86_64.rpm                                                                                                                      63 MB/s | 6.0 MB     00:00
(8/18): mariadb105-gssapi-server-10.5.29-1.amzn2023.0.1.x86_64.rpm                                                                                                              278 kB/s |  15 kB     00:00
(9/18): mysql-selinux-1.0.4-2.amzn2023.0.3.noarch.rpm                                                                                                                           1.4 MB/s |  36 kB     00:00
(10/18): mariadb105-server-utils-10.5.29-1.amzn2023.0.1.x86_64.rpm                                                                                                              6.6 MB/s | 207 kB     00:00
(11/18): perl-DBD-MariaDB-1.22-1.amzn2023.0.4.x86_64.rpm                                                                                                                        8.0 MB/s | 153 kB     00:00
(12/18): perl-DBI-1.643-7.amzn2023.0.3.x86_64.rpm                                                                                                                               19 MB/s | 700 kB     00:00
(13/18): perl-File-Copy-2.34-477.amzn2023.0.7.noarch.rpm                                                                                                                        872 kB/s |  20 kB     00:00
(14/18): mariadb105-server-10.5.29-1.amzn2023.0.1.x86_64.rpm                                                                                                                     66 MB/s |  10 MB     00:00
(15/18): perl-Math-BigRat-0.2624-500.amzn2023.0.2.noarch.rpm                                                                                                                    899 kB/s |  42 kB     00:00
(16/18): perl-Math-BigInt-1.9998.39-2.amzn2023.0.2.noarch.rpm                                                                                                                    3.6 MB/s | 202 kB     00:00
(17/18): perl-Math-Complex-1.59-477.amzn2023.0.7.noarch.rpm                                                                                                                      1.7 MB/s |  46 kB     00:00
(18/18): perl-Sys-Hostname-1.23-477.amzn2023.0.7.x86_64.rpm                                                                                                                      613 kB/s |  16 kB     00:00
----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------
Total                                                                                                                                                                            60 MB/s |  19 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :  [==================================================================]                                                                                                                   ] 1/1
```

systemctl start mariadb

systemctl enable mariadb

```
Complete!
[root@ip-172-31-17-175 ~]# systemctl start mariadb
systemctl enable mariadb
Created symlink /etc/systemd/system/mysql.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/mysqld.service → /usr/lib/systemd/system/mariadb.service.
Created symlink /etc/systemd/system/multi-user.target.wants/mariadb.service → /usr/lib/systemd/system/mariadb.service.
[root@ip-172-31-17-175 ~]#
```

Add second connection step

mysql -h database-1.ckvwacieqq64.us-east-1.rds.amazonaws.com -P 3306 -u admin -p --ssl-verify-server-cert --ssl-ca=./global-bundle.pem

Enter your password which you have given at the time of creating RDS

```
[root@ip-172-31-17-175 ~]# mysql -h database-1.ckvwacieqq64.us-east-1.rds.amazonaws.com -P 3306 -u admin -p --ssl-verify-server-cert --ssl-ca=./global-bundle.pem
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 90
Server version: 11.8.6-MariaDB-log managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

You Enter in the SQL now create database

CREATE DATABASE college;

SHOW DATABASES;

```
MariaDB [(none)]> CREATE DATABASE co
Query OK, 1 row affected (0.008 sec)

MariaDB [(none)]> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| college            |
| information_schema |
| innodb             |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
6 rows in set (0.006 sec)
```

USE college;

Create table:-

```
CREATE TABLE students (
  id INT PRIMARY KEY AUTO_INCREMENT,
  name VARCHAR(50),
  marks FLOAT
);
```

Insert Data:-

```
INSERT INTO students (name, marks)
VALUES ('Suyash', 85);
```

View data:-

```
SELECT * FROM students;
```

```
MariaDB [college]> SELECT * FROM students;
+----+--------+-------+
| id | name   | marks |
+----+--------+-------+
|  1 | Suyash |    85 |
+----+--------+-------+
1 row in set (0.001 sec)

MariaDB [college]>
```

This project demonstrates the creation and management of a relational database on an EC2 instance using Amazon Web Services and MariaDB. The database was configured on a Linux-based EC2 server, where essential operations such as database creation, table design, and data manipulation were performed using SQL.

The project includes implementation of core database functionalities like creating tables, inserting records, retrieving data, and executing queries to analyze information. It also involves basic server management tasks such as installing database software, starting services, and connecting remotely via SSH.

Overall, this project provides hands-on experience with cloud computing and database management, helping in understanding how real-world applications store and manage data efficiently in a cloud environment.